Whose Perception of Geopolitical Risk Is Priced in Defence Equities?

Evidence from the Russia–Ukraine War

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Abstract

Standard measures of geopolitical risk are constructed from English-language newspapers, which makes the editorial vantage point of those measures an implicit modelling assumption rather than a tested one. This thesis asks whose perception of geopolitical risk is priced in defence equities. Using 4,027 days of multilingual news from GDELT's translingual archive, classified by the national identity of the *publisher* rather than by the subject of the article, it tests whether Ukrainian, Russian state and Russian independent coverage of the Russia–Ukraine war explains defence-equity returns conditional on Western coverage. It does not. Local-language perception adds no explanatory power in coverage volume, in tone, or in the anticipated-versus-realised structure of coverage, across two asset classes — defence equities and European natural gas — and one non-market outcome, realised conflict escalation. The design is nonetheless sensitive: the Western block is detected under multiple-testing correction in the pre-registered anticipation test, and local coverage is itself detected in seven of 31 specifications when credited to its publication day, and in one when the same coverage is lagged one trading day so that it is wholly available before the market opens. A power simulation gives the design 81% power to detect a true out-of-sample R^2 of 0.5% on the three long-sample targets, and 44% on the two Bloomberg indices, whose shorter span leaves the null on them appreciably weaker. Descriptively, Russian state media's coverage tone did not change at the full-scale invasion of February 2022 (+0.02) while Ukrainian media's fell by 1.66 points, indicating that the signal which moved most sharply at the defining event of the sample is not the one defence-equity prices track.

1 Introduction

The Russia–Ukraine war is the most intensively covered armed conflict in modern media history, and it produced the sharpest repricing of defence equities in decades. European defence equities rose 21.6% in the five weeks following Russia's full-scale invasion of Ukraine on 24 February 2022, against 7.0% for the global sector index and 9.1% for its U.S. counterpart; the sector then re-rated far more sharply in the first half of 2025 (Europe +63.7%) as European governments announced rearmament budgets. The question this raises is what information drove that repricing. The answer offered by the geopolitical-risk literature is newspaper coverage; which newspapers is left unspecified. The two most widely used indices, those of Caldara and Iacoviello (2022) and Baker et al. (2016), are built from a fixed panel of English-language, mostly American and British outlets. When a defence-equity price is said to be responding to geopolitical risk, it is responding to a quantity constructed from one particular editorial perspective — the Western one.

The research question of this thesis is: whose perception of geopolitical risk is priced in defence equities? Two candidate answers are possible. Either the information that moves defence-equity prices originates where the war is fought — in Ukrainian and Russian media, which are closer to the battlefield, cover it more intensively, and carry it earlier — and reaches Western prices with the delay that translation and attention impose. Or local coverage, however much earlier or more voluminous, adds nothing to Western defence-equity prices once the Western signal is controlled for. The second is the alternative this thesis can test directly, since it is a statement about incremental information; establishing the positive claim that Western coverage is itself priced would require a different design.

The first challenge is measurement, and it is more severe than it appears. Comparing national media ecosystems requires a corpus that actually contains them. GDELT's Global Knowledge Graph version 1.0, the stream most readily available for this purpose, does not: a representative daily file contains 60,690 records, of which 7 carry a `.ru` domain and 21 a `.ua` domain. Nationality in that stream can be assigned only from the countries an article *mentions*, which for 88.6% of articles is not the country that published it. Series built on that basis and labelled Ukrainian or Russian sentiment therefore measure the tone of English-language writing *about* Ukraine and Russia — one editorial population sorted by topic and relabelled as three perspectives — and any comparison between them is a comparison of two topic proxies rather than of two vantage points. This thesis instead uses GDELT's *translingual* archive, which processes material in 65 source languages and identifies each article by the outlet that published it. The resulting indices begin on 18 February 2015, the first day of that archive, and cover 4,027 days.

The second challenge is identification. News coverage of a conflict and defence-equity prices may move together for many reasons that have nothing to do with the media ecosystem carrying the signal. Both respond to the same underlying events; financial market conditions affect both; and a common regime trend contaminates any level regression. The strategy addresses this through a conditional joint F-test: local media are asked whether they add to returns *conditional on Western coverage already being in the regression*. This mirrors the test in Bondarenko et al. (2024): they ask whether local Russian risk measures add to English-language measures for the Russian economy; this thesis asks the same question for the counterparty's assets. The market control must also be regional: European defence returns are controlled with the STOXX 600 rather than the S&P 500, a choice that Section 4.4 shows to be decisive rather than cosmetic.

The third challenge is multiple testing. The specification grid crosses two frequencies, six conflict-episode windows plus the full sample, and five defence-equity targets, of which 31 cells carry enough observations to estimate. Running 31 tests at the 5% level would be expected to produce roughly 1.5 false rejections by chance alone. Benjamini–Hochberg correction is therefore applied across the full grid. Four tests are reported. They are numbered as in the pre-registration series, which begins with a first gate on index construction rather than on the research question, so the tests reported here are Gates 2 to 5: Gate 2 tests coverage volume and tone, Gate 3 the anticipated/realised structure, Gate 4 European natural gas, and Gate 5 realised conflict escalation. Gates 3 to 5 are pre-registered; Gate 2 is not. For Gate 3 the pass rule, the excluded targets and the required sign condition were fixed before the test was constructed. For Gates 4 and 5 the data were collected only after the pre-registration was committed, so in those two cases the ordering of specification and observation is verifiable from the timestamps of the committed pre-registration documents rather than asserted. Gate 2 is reported as the exploratory test it is, and the results below turn on a timing contrast within it rather than on any single cell clearing a threshold.

The fourth challenge is evaluating a null result. A finding that nothing predicts is uninformative unless the magnitude of effect the design could have detected is also established. The out-of-sample section of this thesis therefore accompanies its zero Clark–West rejections with a power simulation: using the actual return series and implanting a predictor with a known true effect, the design has 81% power against a true out-of-sample R^2 of 0.5% and 98% against one of 1.0%, on the 1,855 out-of-sample days available for the three long-sample targets. Power on the two Bloomberg indices, which contribute only 686 out-of-sample days, is materially lower at 44% and 76%. An effect in the range this literature treats as economically meaningful would therefore probably have been detected on the long-sample targets, which is what makes the null informative there; it is not the same as showing that no such effect exists.

The principal finding is that local-language perception — the "local block" of Ukrainian, Russian state and Russian independent indices, tested jointly against the "Western block" of Western and native-English ones — adds nothing to defence-equity returns once Western coverage is in the regression, and that the design is nonetheless able to detect media effects where they exist. The second half of that statement is what makes the first a finding rather than an absence, and it is established test by test rather than in aggregate. The Western block survives multiple-testing correction in the pre-registered anticipation test (2 of 31 cells, minimum $p = 1.6 \times 10^{-4}$), and is detected at $p = 4 \times 10^{-5}$ in the natural-gas test. In the attention-and-tone test the Western block does not survive correction at the tradeable one-day lag and the local block retains a single surviving cell of 31; what demonstrates sensitivity there is the local block's own behaviour across the two alignments: seven of 31 cells at the same-day alignment, one at the conservative lagged alignment on which the verdicts rest.

The principal descriptive finding is an asymmetry in how the two sides' media responded to the invasion itself. Russian state media's coverage tone did not move when Russia invaded Ukraine — a shift of +0.02 in aggregate, and −0.05 on a fixed panel of the 24 state outlets present on both sides of the event — while Ukrainian media's tone fell by 1.66 points on the same scale. This is the largest contrast in the corpus, and the one that survives the composition and break-date checks reported with Table 3, and it is consistent with the pricing result rather than separate from it: the information set that changed most sharply at the defining event of the sample is not the one whose movements defence-equity prices track.

The null extends from contemporaneous association to forecasting. Across 50 out-of-sample specifications the Clark–West test rejects in zero cases against 2.5 expected by chance, the best out-of-sample R^2 is +0.10%, and no specification reaches even nominal significance. The accompanying power simulation establishes that this reflects the data rather than the design.

The primary contribution is to the literature on whose information is priced in financial markets. Bondarenko et al. (2024) show that for the Russian domestic economy, local Russian-language risk perception matters and English-language perception does not. This thesis runs the mirror of that test on the counterparty's assets and finds the asymmetry reversed: local-language coverage carries no incremental information for Western defence-equity returns once Western coverage is in the regression. The symmetric positive claim does not follow, and is not made here. Establishing that Western coverage *is* priced would require it to be detected consistently rather than in some tests and not others, and the evidence falls short of that. What the two studies support jointly is weaker and still useful: the relevance of a media ecosystem to an asset appears to depend on whether its readers are that asset's investors, and neither result licenses a general claim that local media always or never matter.

The second contribution is to the literature that measures risk from newspaper text, where the question of *which* newspapers has been left largely unexamined. Since Baker et al. (2016) and Caldara and Iacoviello (2022) established normalised newspaper counts as a workable proxy for uncertainty and geopolitical risk, the resulting indices have been treated as measurements of a latent state of the world rather than as measurements taken from a particular vantage point. This thesis makes the vantage point itself the object of study, and shows that the choice is empirically consequential in one direction: adding three local-language vantage points to the Western one contributes nothing that survives correction. The converse experiment is not run here, and the results do not establish that the Western vantage point carries pricing information in its own right.

The third contribution is methodological, and concerns how easily this class of measurement goes wrong. Loughran and McDonald (2011) caution that automated sentiment dictionaries require domain-specific validation, which is why GDELT tone is interpreted here as an indicator of coverage negativity rather than as a direct measure of investor beliefs. The same caution applies one level higher, to the corpus rather than the dictionary. A national sentiment series built from an English-only stream measures topic while appearing to measure provenance, and the resulting error is invisible in the output: the series are well behaved, they correlate with events, and they differ from one another. Section 4.4 reports six further specifications that were significant under defensible initial choices and did not survive stricter ones. Both are reported rather than suppressed, because in a literature whose measurements are constructed rather than observed, the conditions under which a result dissolves are as informative as the result itself.

The fourth contribution is to the empirical literature on defence equities and conflict, which is more equivocal than it is often summarised as being. Event studies find sharp price responses at conflict onsets: Covachev and Fazakas (2024) document abnormal returns of up to 12% on European defence stocks in the days following the February 2022 invasion, and Klomp (2025) finds U.S. defence firms gaining 10–15% after the October 2023 attacks on Israel. Evidence that *continuously measured* geopolitical risk predicts defence returns is considerably weaker. Apergis et al. (2018) find that geopolitical-risk shocks predict the volatility of roughly half of 24 leading defence companies but do not predict their returns, and Zhang et al. (2023) similarly locate the effect of geopolitical risk in volatility rather than in the conditional mean across 32 markets. More generally, Tetlock (2007) and Manela and Moreira (2017) establish that media content carries information for asset prices at short and long horizons respectively. What none of this work does is separate news by the nationality of the outlet producing it, which is the separation this thesis performs.

The wider implication concerns an assumption that is increasingly built into commercial risk measurement. Multilingual geopolitical-risk indicators are constructed on the premise that sources closer to an event carry information that Anglophone coverage omits. That premise is testable, and the setting studied here is close to the most favourable one available for testing it: an eleven-year sample, the most intensively covered conflict in modern media history, an asset class whose valuation rests on a claim about that conflict, and local media operating at near-saturation coverage of it. The premise does not hold at either of the horizons at which the information could have been traded. In this setting, proximity to an event does not imply proximity to the price that responds to it.

2 Data and Identification Strategy

2.1 The news corpus and its construction

The analysis rests on a corpus of 4,027 days of conflict-related news drawn from the Translingual Global Knowledge Graph of GDELT (Leetaru and Schrodt, 2013), covering 18 February 2015, the first day of that archive, to 20 May 2026 — 98% of the calendar days in that span. **The sample opens during the war, not before it.** Russia's war against Ukraine began in 2014, with the annexation of Crimea and the onset of fighting in the Donbas, so every observation used here is a wartime observation and the event the sample straddles is the escalation of 24 February 2022 rather than the outbreak of hostilities. This matters for reading the descriptive evidence: the local ecosystems already devote most of their output to articles naming Ukrainian or Russian places in 2015, both because the conflict was already under way and because such places are where their domestic news happens, which is why they have so little room to react when the full-scale invasion comes. Of those 4,027 ingested days, 3,073 carry all five ecosystem series simultaneously and form the estimation sample; the remainder come from a held-out window ingested separately and used only where the text says so. The corpus is accessed through BigQuery's `gdelt-bq.gdeltv2.gkg_partitioned` table, a day-partitioned store of GDELT's global news annotations running to several terabytes, from which only the columns and days required are scanned. **The corpus is not filtered by topic.** Every article an ecosystem published on a given day enters the aggregation, and geolocation to Ukraine or Russia is recorded as a flag on each article rather than used as a condition for inclusion. This distinction determines what the coverage-share variable means, and Section 2.2 sets out both halves of the ratio explicitly. GDELT's translingual stream processes text in 65 source languages, machine-translates it, and applies a unified annotation pipeline, making it possible to compare Ukrainian, Russian, and Western outlets through a single measurement framework.

Each article is assigned to one of five media ecosystems based on the outlet that published it — not based on the country the article discusses. This distinction is critical: an article in a British newspaper about Russian troop movements is Western, not Russian. The assignment uses four tiers in priority order. First, a hand-curated register assigns high-volume outlets manually, verified by country of registration and editorial ownership. Second, outlets with a nationally registered domain suffix (`.ua`, `.ru`, `.de`, etc.) are assigned by ccTLD. Third, source language is used conditional on GDELT country information. Fourth, unresolved articles fall to a residual "other" category. Country of publication takes priority over article language at every tier: Ukrainian outlets publish heavily in Russian (e.g. `24tv.ua` carries 2,595 Ukrainian-language and 1,865 Russian-language articles), and classifying them by language would file those articles as Russian and manufacture agreement between two ecosystems the design needs to keep apart. A further rule governs the two cases in which ownership and location diverge, and it is stated explicitly because it must point in both directions to be a rule at all: state-funded external broadcasters are classified to the funding state, which places Deutsche Welle and Radio Free Europe/Radio Liberty in the Western block, while exile newsrooms are classified to their country of origin, which keeps *Meduza* in the Russian independent block. Section 4.4 shows that misapplying this rule to a small number of high-volume outlets is sufficient to generate a significant result where none exists.

The five resulting ecosystems are: **UA** (Ukrainian outlets), **RU_STATE** (Russian state-controlled), **RU_INDEP** (Russian independent), **WEST** (EU, UK, US, Canada, Australia), and **EN_GLOBAL** (English-language global sources, labelled *native-English* in the tables and figures). Russian state outlets are identified through ownership registers following the press-freedom classification in Bondarenko et al. (2024). Russian independent outlets include platforms exiled or blocked in Russia such as *Meduza* and *TV Rain*.

The register is validated against an external source rather than asserted. Each registered domain is resolved against Wikidata and its recorded country of origin compared with the assigned ecosystem: of 84 registered outlets, 66 resolve to a verifiable country, and 63 of those agree, a precision of **0.955**. The Russian state, Western and Ukrainian blocks agree perfectly (1.000 each); all three disagreements fall in the Russian independent block (0.750), and all three are exile newsrooms, which is the one case where country of origin and country of operation genuinely differ. This audit validates that each registered domain belongs to the country it was assigned to. It does not validate GDELT's attribution of a given article *to* a domain, which is taken on trust and would require a reader with Russian and Ukrainian to check.

The classification rule is also varied, to establish that the results are not an artefact of it. Five rules are run through the full test grid: the baseline, a register-only rule that drops the ccTLD and language tiers, a rule with no language tier, a language-first rule, and one that retains news aggregators. The local-block verdict is unchanged under all five — one or two survivors of 31 cells under the primary alignment in every variant. The language-first rule is the informative failure: because the language tier claims Russian-language articles before ownership is consulted, it cannot represent a Russian independent block at all, which is the concrete cost of the country-before-language rule being got wrong.

2.2 Perception indices

Two indices are constructed for each ecosystem per day, and because their interpretation turns on it, both halves of the ratio are stated explicitly.

The denominator is the ecosystem's entire output, not a filtered subset. It is a count of every article published that day by an outlet assigned to the ecosystem, with no topical restriction: the only conditions applied are the date partition and a non-missing source name. Numerator and denominator are computed in the same server-side aggregation, as `COUNTIF(is_conflict)` over `COUNT(*)` within each day–ecosystem group, so the denominator cannot drift out of step with the numerator. Its magnitude confirms that it is unfiltered: on a median day it is about 6,400 articles for the Ukrainian ecosystem and 115,000 for the Western one, which is the scale of those ecosystems' total production rather than of their conflict reporting. Using a share rather than a raw count is standard practice (Baker et al., 2016), and here it is necessary: GDELT's source coverage drifts by a factor of two and a half over eleven years, which a count series would register as a trend in every ecosystem at once.

The numerator is broader than the word "conflict" suggests, and the variable is named accordingly. An article counts if its GDELT location field contains a place in Ukraine or Russia. That is a geographic criterion, not a topical one, and it was chosen because the location field costs 0.242 TB to scan over the full sample against 1.853 TB for the theme field — the difference between a query inside the free tier and one outside it. The variable is therefore called the **Ukraine/Russia coverage share** rather than a conflict-attention share, and the **coverage tone** is the mean GDELT tone over those same articles, on a dictionary-based scale running from about -10 to $+10$.

The consequence is that this proxy is sharper for some ecosystems than for others, and the asymmetry runs against the local blocks. For a Western outlet, mentioning a Ukrainian or Russian location is unusual and largely implies conflict coverage: the median Western day is 5.9% such articles. For a Ukrainian outlet it implies very little, because domestic politics, business, sport and crime all mention Ukrainian places; the median Ukrainian day is 80.2%, and the Russian ecosystems sit at 70% and 86%. The local series are consequently close to a ceiling imposed by the measure itself, which limits the headroom in which an escalation could show up as a change in coverage share. Section 5 returns to this as a limitation. It does not affect the anticipated/realised variables introduced next, which are built from GDELT's theme codes and are conflict-specific by construction.

Both indices are used in daily first differences in all regressions. The reason is empirical: in levels, all series carry a common "war regime" trend that financial controls already capture, and it is the day-to-day *innovation* in perceived risk, not its persistent level, that should carry price-relevant information. The first-difference transformation is also what Bondarenko et al. (2024) apply.

Following Caldara and Iacoviello (2022), each ecosystem's Ukraine/Russia coverage is further split, as a share of that ecosystem's Ukraine/Russia articles rather than of its total output, into an **anticipatory component** (articles classified by GDELT's Themes field as covering threats, build-ups, or anticipated hostilities) and a **realised component** (articles covering acts of war, escalation, or attacks). This threat/act split is the object of the anticipation test in Section 4.1.

2.3 Financial targets and controls

The primary financial targets are two official Bloomberg defence-equity indices: the Bloomberg World Large, Mid and Small Aerospace & Defense Total Return Index (**WAERLST**, USD) and the Bloomberg Europe Defense Select Total Return Index (**BSHIELDT**, EUR), both running from 2 January 2020 to 29 June 2026 (1,619 trading days, no missing observations). Three further targets extend coverage backwards and separate the two regions, giving five in total: the iShares U.S. Aerospace & Defense ETF (**ITA**), available for the whole 2015–2026 window and so covering the years before the full-scale invasion that the Bloomberg indices miss; and two hand-built equal-weighted baskets of listed defence names, one European (**EU basket**) and one American (**US basket**), which extend the same coverage to constituents rather than index vehicles. The two baskets are constructed from free data and are the targets the Gate 3 pre-registration excluded from its primary arm, precisely because they are hand-built.

The choice of market control is not a matter of convenience, and Section 4.4 shows it to be the specification decision on which the largest results in this setting turn. Controlling *European* defence returns with the S&P 500 is superficially reasonable, since the two markets are highly correlated over long horizons. Over the build-up window, however, their correlation is only 0.409, so a U.S. control leaves a substantial component of ordinary European market variation in the residual — and that residual correlates 0.26 with the threat measure being tested. Every regression reported here therefore uses the regional market return appropriate to its target: the STOXX 600 for European targets and the S&P 500 for global and U.S. ones. The only other control is the VIX, which absorbs general risk appetite, and it enters as the natural logarithm of the previous day's close, standardised. Lagging it matters: the same-day VIX is contemporaneous with the return being explained and would absorb part of what the news variables are being asked to account for. The defence-equity specifications contain no lagged dependent variable. Gate 4 adds one, because daily gas returns are materially autocorrelated where defence returns are not, and Gate 5 uses no market control at all, since no equity index is the relevant benchmark for a geopolitical-risk series, conditioning instead on its own dynamics. Both departures are noted where they arise.

2.4 Physical attack data (supplementary context)

The analysis is grounded in media perception rather than physical events, but a parallel dataset records the physical dimension of the conflict: daily Ukrainian Air Force records of Russian air attacks from 29 September 2022 to 21 June 2026. The raw file contains 3,812 attack-level records, aggregated to 809 unique market-information dates using the rule

$$\text{market\_info\_date} = \max(\text{attack\_date},\ \text{time\_end\_date}), \tag{1}$$

which assigns each attack to the day on which the verified count could plausibly enter the market's information set, avoiding look-ahead bias. Across all records, 102,396 weapons were launched and 76,126 were intercepted or destroyed (interception rate: 74.3%); UAVs account for 94.8% of all launched weapons. War intensity is measured as $\ln(1 + \text{weapons launched}_t)$.

This dataset is used in two ways. First, it provides physical context for the media patterns in Section 3: Western news attention spikes on high-intensity attack days, while local coverage share is already near saturation. Second, it enters the forecasting comparison in Section 4.4 as a competing information set, where realised attack intensity turns out to be no more informative about subsequent defence returns than the news measures are.

The dataset binds the joint sample to September 2022 onward, which is why the primary analysis uses GDELT-only over the longer 2015–2026 window. The physical and perceptual dimensions of the conflict are thus separated in the sample design rather than conflated.

2.5 Identification strategy

The core test asks whether local media add explanatory power to defence returns *conditional on* the Western signal. The specification for each cell in the grid is:

$$r_{i,t} = \alpha + \beta_{\text{West}} \mathbf{X}_t^{\text{West}} + \beta_{\text{Local}} \mathbf{X}_t^{\text{Local}} + \gamma Z_t + \varepsilon_{i,t}, \tag{2}$$

where $r_{i,t}$ is the return on defence-equity target i, $\mathbf{X}_t^{\text{West}}$ contains attention-share and tone changes for the Western and EN_GLOBAL ecosystems, $\mathbf{X}_t^{\text{Local}}$ contains the same for UA, RU_STATE, and RU_INDEP, and Z_t contains exactly two controls: the contemporaneous return on the regional market index, and the standardised log VIX of the previous close. Standard errors are HAC(5) throughout, and the same Z_t is used in Gates 2 and 3 and in Table 4.

The object of interest is a **joint F-test on** β_{Local}, with the Western block already in the regression. This conditioning is what separates "local media add something" from "local and Western media correlate with returns for the same reason". The test is run across frequency (daily and weekly), six conflict-episode windows and the full sample, and five targets. That grid has 70 cells, of which 31 clear a minimum-observation rule requiring at least 40 observations and at least four per estimated parameter. The remainder are too thin to estimate reliably and are dropped before any test is run, rather than being estimated and then discounted afterwards.

Two conventions recur in every results table and are defined here once. "BH survivors" is the number of cells that remain significant after Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) controlling the false discovery rate at 5% across all 31 cells of an arm simultaneously; throughout this thesis it is that count, and not the number of nominally significant cells, that decides a verdict. "HAC(5)" denotes Newey–West standard errors (Newey and West, 1987) with a five-day bandwidth, which remain valid under the serial correlation and heteroskedasticity that daily return regressions exhibit.

The **positive control** is the identical joint F-test on β_{West} in the same 31 cells. If the design detects the Western block, a null on the local block is informative. If it detects neither, the absence of the local result is merely consistent with low power.

The out-of-sample test in Section 4.3 uses an expanding-window design with a minimum 250-day training window, re-estimated at each step and forecasting one day ahead.

The choice of test statistic for that exercise follows from the structure of the comparison and is not interchangeable. Forecast-accuracy comparisons in this literature are most often reported using the Diebold–Mariano statistic (Diebold and Mariano, 1995), which is not valid here. Each of the 50 specifications compares a model — the historical mean augmented with one perception predictor — against the historical mean alone, and setting that predictor's coefficient to zero recovers the benchmark exactly. The benchmark is therefore nested within the model, and under nesting the Diebold–Mariano statistic has no standard normal limiting distribution and is biased against the larger model even when its additional predictor carries genuine information. Clark and West (2007) correct this by adjusting for the estimation noise that nesting introduces, yielding a statistic that is valid in this setting and one-sided in the direction of improvement. Diebold–Mariano with the Harvey et al. (1997) small-sample correction is retained for genuinely non-nested comparisons, but every cell of the reported grid is nested and is arbitrated by Clark–West.

3 Descriptive Statistics

The descriptive evidence establishes two things before any regression is run: that the local and Western media ecosystems are not measuring the same thing, and that the one series which moved most sharply at the invasion belongs to the ecosystem that turns out not to be priced. Table 1 defines every variable used in the analysis and Table 2 reports its distribution; Table 8, in the appendix, reports how the series co-move; Figures 1 and 2 plot the outcome and the regressors over the full sample; and Table 3 isolates each ecosystem's tone response to the invasion, the event with the greatest identifying power in the sample.

Two features of Table 2 carry most of the descriptive argument. The first is in Panel B: Ukrainian, Russian state and Russian independent outlets devote 71–85% of their output to articles naming a Ukrainian or Russian place, against 7.2% for Western outlets and 5.4% for native-English ones. That is an order-of-magnitude difference, sustained across eleven years, and it is the sense in which the local ecosystems are "closer" to the war — though for them the measure largely reflects domestic geography rather than conflict reporting specifically, as Section 2.2 explains. The local series are also far less variable *relative to* those means: Ukrainian attention has a standard deviation of 5.0 percentage points on a mean of 81.0, a coefficient of variation of 0.06, against 0.63 for Western attention. An outlet already devoting four-fifths of its output to a subject has little room left to escalate, which is the mechanical reason the local levels move so little in proportional terms.

The second feature pre-empts the most obvious objection to the results that follow. Panel D reports the daily changes that enter the regressions, and shows that the local series are not inert. Those changes are mean-zero by construction, so it is their dispersion that matters: the daily change in Ukrainian attention has a standard deviation of 2.3 percentage points and that in Russian independent attention 4.1, against 1.2 for Western outlets and 1.0 for native-English ones. The local *attention* regressors therefore vary two to four times as much from day to day as the Western ones, while tone changes are comparably variable across all five ecosystems (standard deviations of 0.37 to 0.42 points). Whatever explains the null in Section 4, it is not an absence of variation in the local regressors.

Figure 1 plots the outcome variable, and shows that the war repriced defence equities in two distinct steps rather than one. The first step is the full-scale invasion itself: in the five weeks from 24 February 2022 the European index gains 21.6%, against 7.0% globally and 9.1% in the United States. The second, and much larger, step comes three years later, when European governments announced rearmament programmes: the European index gains 63.7% in the first half of 2025 alone and ends the sample at roughly 2.5 times its January 2020 level, while the global and U.S. indices end near 1.9 times. Two features of this figure matter for what follows. The European index is the one that responds to the conflict, which is why the regional market control in Section 2.5 is a European rather than a U.S. index. And the largest single move in the sample is driven by European fiscal announcements rather than by battlefield news, which is a reason to test war-news predictability against a market control rather than in isolation.

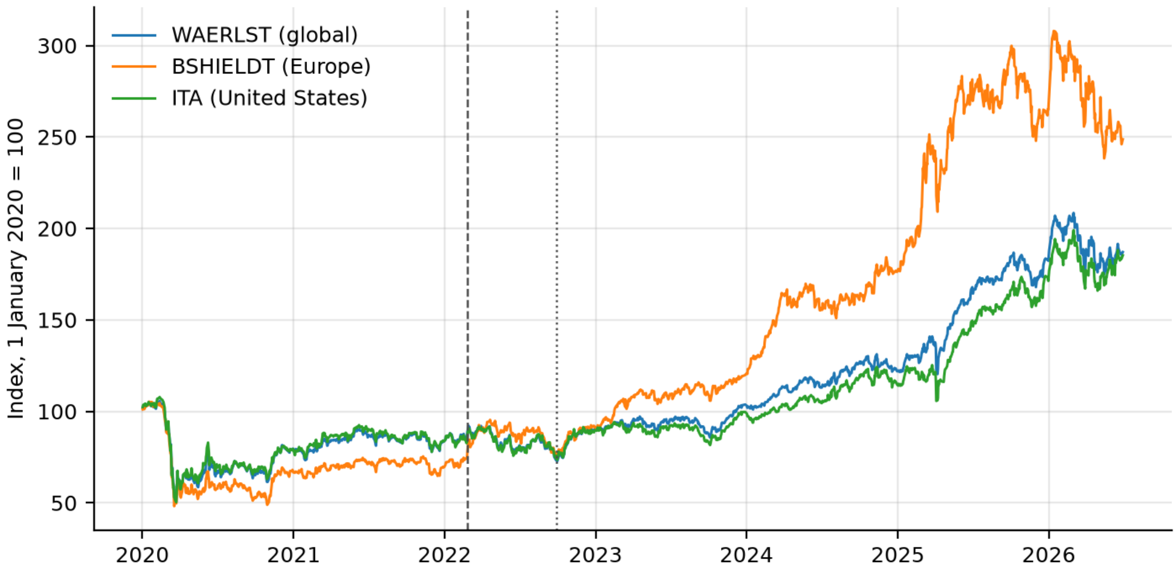


Figure 1: Defence-equity index levels, 1 January 2020 = 100, to June 2026. **WAERLST** (blue) is the Bloomberg World Aerospace & Defense Total Return Index; **BSHIELDT** (orange) is the Bloomberg Europe Defense Select Total Return Index; **ITA** (green) is the iShares U.S. Aerospace & Defense ETF. The dashed vertical line marks the full-scale invasion (24 February 2022); the dotted vertical line marks 29 September 2022, the first day of the Ukrainian Air Force attack record used in Section 2.4. All three series are total-return indices, so the vertical distance between them is cumulative performance including dividends, not a price-only comparison.

Figure 2 plots the perception indices that are the regressors of interest. The dominant feature of Panel A is that the two media environments occupy entirely different levels throughout the eleven years. Ukrainian and Russian outlets devote 71–85% of their output

Table 1: Variable definitions

Variable	Unit	Definition
Financial outcomes		
WAERLST return	%	Daily log return on the Bloomberg World Aerospace & Defense Total Return Index.
BSHIELDT return	%	Daily log return on Bloomberg Europe Defense Select Total Return Index.
ITA return	%	Daily log return on iShares U.S. Aerospace & Defense ETF.
Perception indices (one per ecosystem per day; full 2015–2026 sample)		
Ukraine/Russia coverage share	%	Articles whose GDELT location field names a place in Ukraine or Russia, as a share of *all* articles the ecosystem published that day.
Coverage tone	score	Mean GDELT tone of those same articles (range ≈ -10 to $+10$; more negative = more negative coverage).
Anticipatory share	%	Articles classified by GDELT Themes as covering threats or build-ups, as a share of that ecosystem's Ukraine/Russia articles. Theme-based, so conflict-specific. The "threat" half of the Gate 3 split.
Realised share	%	Articles classified as covering acts of war, attacks or escalation, same denominator. Theme-based, so conflict-specific. The "act" half of the Gate 3 split.
Non-market outcome		
GPR_ACT	index	The realised-conflict component of the Caldara and Iacoviello (2022) geopolitical risk index; the outcome in Gate 5, where no asset price is involved.
Physical attack variables (robustness subsample only, Sep 2022 – Jun 2026)		
Attack intensity	log count	$\ln(1 + \text{weapons launched}_t)$ on the market-information date; zero on non-attack days.
Interception rate	ratio	Weapons destroyed divided by weapons launched; captures attack effectiveness.
Financial controls		
Market return	%	STOXX 600 (European targets) or S&P 500 (global/U.S. targets) daily log return.
Log VIX, lagged	index pts	Natural log of the previous day's CBOE Volatility Index close, standardised. The one control besides the market return in the defence-equity regressions.
TTF gas return	%	Daily log return on Dutch TTF month-ahead natural gas; the Gate 4 outcome.

Notes: All perception indices are taken in daily first differences in regressions, for the reason given in Section 2. Ecosystems are UA (Ukrainian), RU_STATE (Russian state-controlled), RU_INDEP (Russian independent), WEST (EU, UK, North America, Australia), EN_GLOBAL (English-language global). See Section 2 for classification details.

Table 2: Descriptive statistics

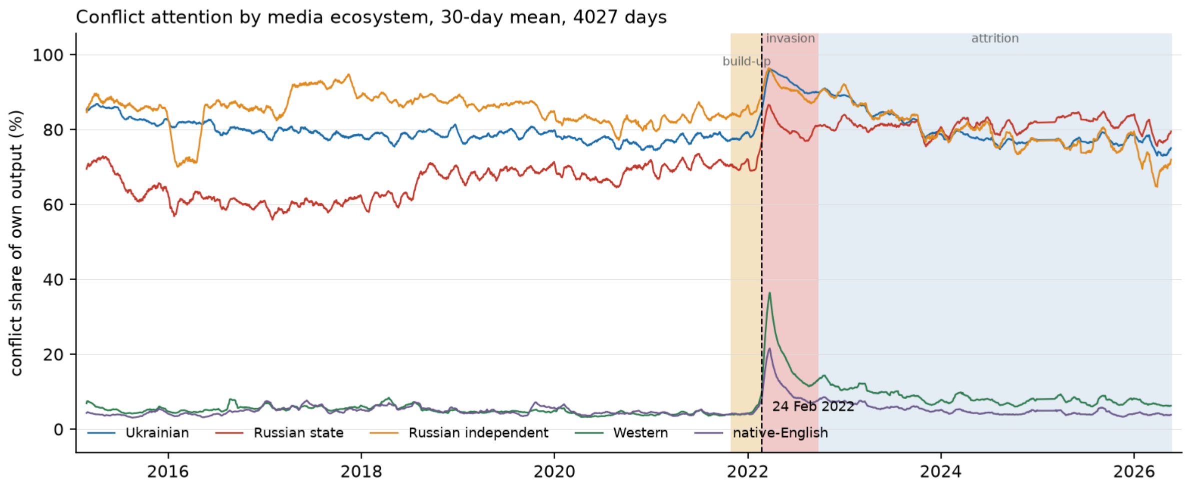
Variable	Mean	Std. dev.	Min	Max	N
Panel A: financial outcomes and controls, daily % returns					
WAERLST return	0.05	1.55	−13.74	10.56	1,619
BSHIELDT return	0.07	1.79	−13.41	12.14	1,619
ITA return	0.05	1.42	−15.95	11.60	2,837
STOXX 600 return	0.02	1.03	−12.19	8.07	2,782
VIX (index points)	18.38	7.05	9.14	82.69	2,837
Panel B: Ukraine/Russia coverage share (% of the ecosystem's daily output)					
Ukrainian	81.0	5.0	60.1	97.1	3,073
Russian state	70.6	9.3	45.4	92.7	3,073
Russian independent	84.8	7.0	41.3	99.7	3,073
Western	7.2	4.5	2.7	47.2	3,073
Native-English	5.4	2.5	2.3	32.2	3,073
Panel C: coverage tone (GDELT score, ≈ −10 to +10)					
Ukrainian	−2.39	0.63	−5.05	0.78	3,073
Russian state	−1.78	0.45	−3.60	2.03	3,073
Russian independent	−2.31	0.65	−5.50	0.54	3,073
Western	−1.80	0.51	−4.01	1.32	3,073
Native-English	−1.99	0.49	−4.20	−0.38	3,073
Panel D: the actual regressors — daily changes in the Panel B and C series					
Δ Ukrainian attention (pp)	0.00	2.3	−12.4	10.8	3,072
Δ Russian state attention (pp)	0.00	3.2	−18.6	21.3	3,072
Δ Russian indep. attention (pp)	0.00	4.1	−28.3	35.5	3,072
Δ Western attention (pp)	0.00	1.2	−10.4	21.9	3,072
Δ Native-English attention (pp)	0.00	1.0	−7.0	12.0	3,072
Δ Ukrainian tone (points)	0.00	0.42	−2.08	1.84	3,072
Δ Russian state tone (points)	0.00	0.37	−3.31	2.66	3,072
Δ Western tone (points)	0.00	0.40	−1.93	1.46	3,072
Panel E: physical attacks, attack days only (robustness subsample)					
Weapons launched	126.6	144.2	1	982	809
Mean daily interception rate	0.735	0.211	0.000	1.000	809

Notes: Panels A–D summarise each series over the full span on which it exists: the Bloomberg indices from 2 January 2020, the news series from 18 February 2015 to 20 May 2026. Panel D reports the daily first differences that enter equation (2), attention in percentage points and tone in index points; two of the ten change series are omitted for space. Panel E covers the 809 dates carrying at least one recorded weapon launch.

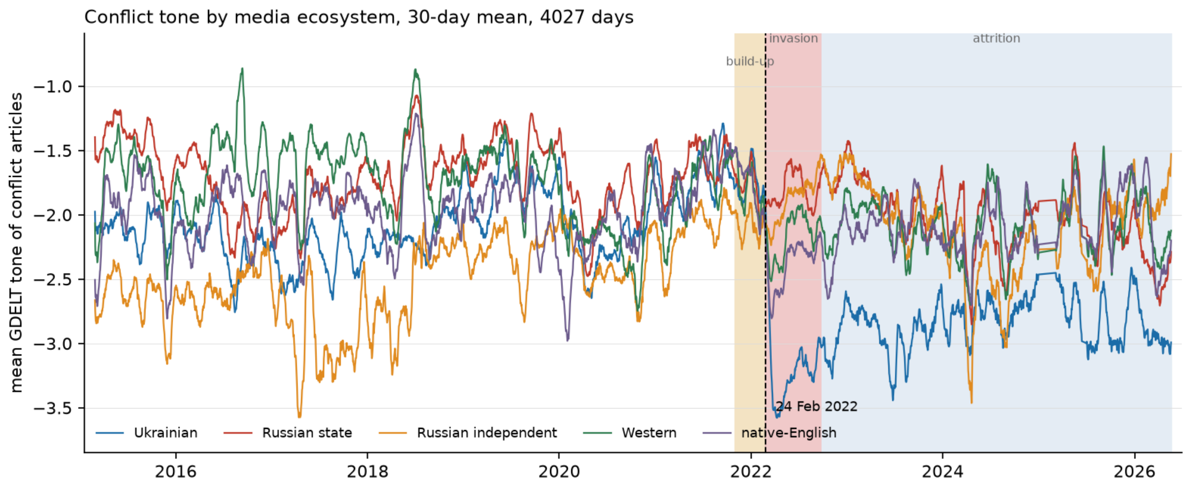
to articles naming a Ukrainian or Russian place and never fall below 41%, while Western and native-English outlets average 7.2% and 5.4% and remain in single digits until the build-up of late 2021. The two ecosystems differ in level by an order of magnitude for almost the whole sample. In the underlying daily series that Panel A smooths, the invasion registers as a one-day step: Western attention moves from 19.0% on 23 February 2022 to 40.9% on the 24th, and native-English attention from 11.8% to 23.8%. In the same two days Ukrainian attention moves from 90.3% to 95.8% and Russian state attention from 84.6% to 91.2%: the local ecosystems had almost no room left to react, which is the mechanical reason a Western investor watching coverage volume would have seen a signal in Western outlets and nothing much in local ones.

Panel B shows an asymmetry at the invasion. Ukrainian coverage tone falls sharply and stays down; Russian state tone barely moves. Before 2022 the Ukrainian, Russian state, Western and native-English series interleave without a stable ordering, with only Russian independent running consistently more negative; after 24 February 2022 the Ukrainian line separates downward, by -1.66 points, and remains separated for the rest of the sample, while the Russian state line stays within the band it occupied before the full-scale invasion.

Russian state media's tone did not change when Russia invaded Ukraine, and Table 3 quantifies how far that is from what every other ecosystem did. Ukrainian tone falls by 1.66 points, native-English by 0.64, Western by 0.36; Russian state tone moves by +0.02, which is to say not at all. The result is not an artefact of which outlets happen to be in the corpus on either side of the invasion: on a fixed panel of the 24 state outlets present in both windows with at least 200 Ukraine/Russia articles in each, the mean shift is -0.05. Several state outlets in fact became *more positive* — *Gazeta.ru* (+0.48), *Ren.tv* (+0.43), *mskagency.ru* (+0.40) — which is what one would expect from outlets covering the launch of an operation their government had announced as a success. This contrast is the largest and most robust descriptive fact in the data, and it is the reason the pricing null that follows is interesting rather than merely negative: the ecosystem whose signal changed most sharply at the defining event of the sample is not the ecosystem whose signal the market tracks.



Panel A. Ukraine/Russia coverage share (% of each ecosystem's own output)



Panel B. Mean tone of that coverage (GDELT score)

Figure 2: Perception indices by media ecosystem, 30-day moving averages, 18 February 2015 to 20 May 2026. In both panels the five lines are **Ukrainian** (blue), **Russian state-controlled** (red), **Russian independent** (orange), **Western** (green) and **native-English** (purple). Shaded bands mark three phases against an unshaded remainder that runs from the start of the sample to late 2021: build-up (tan, November 2021 to 23 February 2022), invasion (pink, 24 February to June 2022) and attrition (blue, July 2022 onward); the dashed vertical line marks 24 February 2022. Panel A plots the percentage of each ecosystem's daily article output whose GDELT location field names a place in Ukraine or Russia, and the gap between the upper three lines and the lower two is its point: local outlets cover the conflict continuously, Western-facing ones only around the invasion. Panel B plots the mean GDELT tone of those same articles, on a scale running from -10 to $+10$; the plotted range of roughly -3.5 to -0.8 reflects the uniformly negative tone of conflict reporting, and the separation of the Ukrainian line after February 2022, against a flat Russian state line, is the asymmetry quantified in Table 3.

Table 3: Conflict tone before and after the full-scale invasion, by ecosystem

Ecosystem	Pre-invasion (Nov 2021–23 Feb 22)	Post-invasion (24 Feb–5 Jun 22)	Shift
Ukrainian (UA)	−1.77	−3.43	**−1.66**
Native-English (EN_GLOBAL)	−1.87	−2.51	−0.64
Western (WEST)	−1.90	−2.27	−0.36
Russian state (RU_STATE)	**−1.81**	**−1.79**	**+0.02**
Russian independent (RU_INDEP)	−2.09	−2.01	+0.09

Notes: Mean GDELT tone of each ecosystem's Ukraine/Russia articles over the two windows in the column headings (115 pre-invasion and 102 post-invasion days), ordered by the size of the shift. The Russian state figure is confirmed on a fixed panel of the 24 state outlets present in both windows with at least 200 Ukraine/Russia articles in each, on which the mean shift is −0.05. A supremum-Wald test for a break at an unknown date places Ukrainian tone's largest break on 24 February 2022 and Russian state tone's 393 days later.

4 Empirical Findings, Interpretation, and Implications

4.1 Gates 2 and 3: local perception and defence-equity returns

Before the grid of tests, Table 4 reports the estimated specification itself, so that the size of the associations at issue is on the page rather than implied by a p-value. Column (1) contains only the two controls of equation (2), the regional market return and the lagged log VIX; column (2) adds the Western block; column (3) adds the local block with news credited to the day it was published; and column (4) is the same full model with the news lagged by one trading day, the only alignment at which it could have been traded. Regressors are standardised, so a coefficient is the percentage-point change in the daily BSHIELDT return associated with a one-standard-deviation move in that series.

Table 5 then asks the same question across the full grid rather than in one cell. Gate 2 asks it of raw coverage volume and tone. Gate 3 asks it of structure, testing the local split between anticipated and realised conflict, which media in a country at war record very differently from media in a country watching one. Gate 2 is exploratory. Gate 3 is pre-registered, with its pass rule, its excluded targets and its sign condition fixed in a committed document before the test was estimated — a distinction that matters here, because Gate 3 initially cleared that rule on the data available at the time and failed it once the held-out window was added.

Table 4: The main specification, estimated

	(1) Controls only	(2) + Western block	(3) + local, same-day	(4) + local, lagged 1 day
Δ Ukrainian attention			0.009	−0.011
			(0.045)	(0.044)
Δ Russian state attention			−0.001	−0.001
			(0.061)	(0.061)
Δ Russian indep. attention			−0.125***	0.010
			(0.042)	(0.049)
Δ Ukrainian tone			−0.073*	−0.038
			(0.043)	(0.036)
Δ Russian state tone			0.127**	−0.055
			(0.054)	(0.053)
Δ Russian indep. tone			0.033	−0.017
			(0.059)	(0.052)
Δ Western attention		0.040	0.025	0.053
		(0.043)	(0.060)	(0.045)
Δ Native-English attention		0.032	0.010	0.030
		(0.071)	(0.063)	(0.071)
Δ Western tone		0.012	−0.070	0.052
		(0.046)	(0.056)	(0.053)
Δ Native-English tone		−0.009	0.034	0.003
		(0.052)	(0.054)	(0.053)
STOXX 600 return	0.925***	0.923***	0.927***	0.921***
	(0.066)	(0.067)	(0.065)	(0.067)
Log VIX (lagged)	0.068	0.066	0.075	0.066
	(0.051)	(0.050)	(0.049)	(0.049)
Observations	921	921	921	921
R^2	0.348	0.351	0.365	0.354
p, local block jointly zero	—	—	**0.003**	**0.734**
p, Western block jointly zero	—	0.333	0.688	0.177

Notes: The dependent variable is the daily percentage return on BSHIELDT, the Bloomberg Europe Defense Select Total Return Index, over the 921 days on which it trades and all five news series exist. All news regressors are daily first differences standardised to unit variance, so a coefficient is the percentage-point return change associated with a one-standard-deviation move. Column (1) contains the controls alone; (2) adds the Western block; (3) adds the local block with news credited to its publication day; (4) is the same full model with news lagged one trading day. Newey–West HAC(5) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Local-media joint F-tests on defence-equity returns (Gates 2 and 3)

Gate	Block and news timing	Cells	BH surv.	Min. p	Verdict
Gate 2: coverage volume and tone (exploratory)					
2	Local, news lagged 1 day *(primary)*	31	1	9.0×10^{-4}	not robust[a]
2	Local, same-day	31	**7**	$\mathbf{2.2 \times 10^{-6}}$	—[b]
2	*Western, news lagged 1 day (control)*	31	0	2.7×10^{-3}	not detected
2	*Western, same-day (control)*	31	0	1.8×10^{-2}	not detected
Gate 3: anticipated/realised split (pre-registered)					
3	Local, news lagged 1 day *(primary arm)*	31	2	5.5×10^{-6}	FAIL[c]
3	Local, same-day *(secondary arm)*	31	0	—	FAIL
3	*Western, news lagged 1 day (control)*	31	**2**	$\mathbf{1.6 \times 10^{-4}}$	detected

Notes: Each cell is a joint F-test on one block of coefficients in equation (2), with the other block, the regional market control and VIX already in the regression; HAC(5) standard errors. The grid crosses two frequencies, six conflict-episode windows plus the full sample, and five targets, of which 31 cells clear the minimum-observation rule. Benjamini–Hochberg correction is applied across all 31 cells of an arm at a 5% false discovery rate; "BH surv." is the resulting count and "Min. p" the smallest uncorrected p-value in the arm. The lagged alignment is primary because it is conservative: a GDELT day is a full UTC day while European markets close at 16:30 UTC, so a same-day aggregate contains coverage published after the return it is matched to, whereas a lagged one is entirely available before the open. It does not follow that the lag isolates post-close information — it shifts the whole daily bundle — and daily aggregates cannot resolve intraday timing. Verdicts refer to the block under test; control rows report only whether the Western block is detected. **The two gates are held to different standards, because only one of them set a standard in advance.** Gate 3 was pre-registered and either meets its stated pass rule or does not, so its rows read FAIL. Gate 2 is exploratory and had no such rule, so its rows report what was found rather than a verdict against a threshold it never had.

[a] One cell of 31 survives correction. It is the 2017–19 episode window on ITA at weekly frequency, years before the full-scale invasion, and no cell in any window contemporaneous with the events the thesis concerns survives.

[b] Reported for comparison with the row above rather than as a result. Under this alignment the regressor includes coverage published after the return it is matched to, so a detection here is not evidence of a pricing relationship.

[c] Neither pre-registered arm clears. The first required a surviving cell in the Russia build-up window on a Bloomberg target or ITA, and that window produces none, its smallest p being 0.018. The second required same-sign survivors in two independent episode windows, and the two survivors share one. Both survivors are weekly cells in the 2025–26 window fitting 13 parameters to 58 observations, while the two deepest cells in the grid, at 2,754 observations each, return $p = 0.08$ and $p = 0.36$.

The timing contrast in Gate 2 is what makes low power an implausible explanation of the null. Local coverage is strongly associated with defence returns when credited to its publication day: seven of 31 cells survive correction, the strongest at $p = 2.2 \times 10^{-6}$.

Credited instead to the following trading day, seven survivors become one — and that one falls in the 2017–19 window, years before the full-scale invasion and during the low-intensity phase of the conflict, where neither the coverage volumes nor the price movements this thesis is about occur.

The two alignments are not equally interpretable, and the lagged one is preferred because it is conservative rather than because it isolates anything. A GDELT day is a full UTC day, while European markets close at 16:30 UTC, so a same-day aggregate mixes coverage published before the close, which the market could have traded on, with coverage published after it, which it could not. A same-day association is therefore not interpretable as a pricing relationship: part of the regressor postdates the return it is being related to. Lagging the aggregate by one trading day removes that contamination, since every article in it precedes the market open. What it does not do is separate the two components. The lag shifts the entire daily bundle rather than trimming the post-close portion from it, so it discards the genuinely tradeable pre-close coverage along with the rest.

That distinction bounds what the collapse from seven survivors to one can be said to show. It is consistent with same-day news being impounded within the trading day, with the same-day association being a mechanical artefact of aligning a return with coverage partly published after it, and with genuine information that decays inside twenty-four hours. **Full-day aggregates cannot cleanly separate these, because they cannot resolve timing within the day at all.** What the contrast does establish is narrower and still sufficient: the design detects the local block readily under the permissive alignment, so the null under the conservative one is not a failure to measure. Distinguishing the explanations would require intraday timestamps, which Section 5 records as the extension most likely to settle it.

Gate 3 reaches the same verdict on the structural version of the question, and its two survivors fail the pre-registered rule for an instructive reason. They are both weekly cells in the 2025–26 episode window, fitting 13 parameters to 58 observations, while the two deepest cells in the grid — pooled daily specifications with 2,754 observations each — return $p = 0.08$ and $p = 0.36$. An effect that concentrates in the thinnest corner of a grid and disappears where the data are deepest is the signature of noise rather than of a small true effect awaiting more power. Neither pre-registered arm clears in any case. The first required a surviving cell in the Russia build-up window on a Bloomberg target or ITA, and that window produces no surviving cell at all, its smallest p being 0.018. The second required same-sign survivors in two independent episode windows, and the two that survive share a window. The Western block, tested in the same 31 cells under the same correction, does survive, at a minimum p of 1.6×10^{-4}. That asymmetry — the Western block detected and the local block not, in identical cells — is what the test was built to isolate.

Gate 3 also produced a coefficient pattern with an economic interpretation, which a held-out test did not support. Among the surviving weekly cells the Ukrainian coefficients were consistent with "buy the rumour, sell the fact": a shift toward anticipatory coverage raised defence returns and a shift toward realised conflict lowered them. Those four sign predictions were recorded and then tested on the 2017–19 window, which had not been examined when the hypothesis was formed, across the three defence-equity targets available there — twelve sign comparisons in all. Seven matched, against six expected by chance (binomial $p = 0.387$). The out-of-sample signs are indistinguishable from chance. The episode illustrates a general point about pre-registration in this setting: a test fixed in advance but estimated on a sample that is still being extended is not, in the relevant sense, pre-registered, because the data that would discipline it have not yet been observed.

4.2 Gates 4 and 5: gas and escalation

Gates 4 and 5 move the test to the two outcomes where local information should matter most, and the null survives both. If the defence-equity result were an artefact of defence equities being a poor testbed — which is a fair objection, since the link from a Russian newspaper to a U.S. contractor's share price runs entirely through Western investors — then it should break on an asset with a direct physical transmission channel, or on an outcome with no market in it at all. Table 6 tests both. European natural gas is the asset through which the conflict physically transmitted to Europe; realised escalation is the war outcome itself, with no asset price involved. Both tests were pre-registered before their data were ingested.

Gate 4 shows that the null is not a property of defence equities being a weak testbed. European gas is the asset where Russian supply signals should matter most — Russia supplied roughly 40% of EU gas and TTF moved from about €20 to over €300 — and Russian state media is the channel through which supply intent was signalled. Yet the local block returns $p = 0.399$ in the build-up window, and all four pre-registered conditions fail, in four different ways. There is no surviving cell in either required window. The placebo condition is breached, since local perception is jointly significant for Brent crude ($p = 0.0014$) as readily as for European gas, and Brent is an asset the supply-signalling mechanism does not predict. The build-up estimate does not survive dropping the ten largest TTF moves, going from 0.399 to 0.840. And the ordering is inverted: Russian *independent* rather than Russian *state* media leads the local block, which is the wrong way round for a channel that is supposed to run through state signalling of supply intent. In the same specification the Western block is detected at $p = 4{\times}10^{-5}$, which demonstrates

Table 6: Local-media joint tests on European gas (Gate 4) and realised escalation (Gate 5)

Gate / Window	n	p (local block)	p (Western block)
Gate 4: Dutch TTF natural gas returns			
(a) Build-up and invasion, daily	222	0.399	**0.00004**
(b) Shutdown and aftermath, daily	231	0.533	0.193
(c) Full crisis, daily	563	0.250	0.066
(c) Full crisis, weekly	133	0.907	0.052
Gate 5: realised escalation (GPR_ACT), held-out sample			
Horizon $h = 1$ day	637	0.212	0.171
Horizon $h = 5$ days	629	0.227	**0.033**

Notes: Each entry is the p-value of a joint F-test on one block, with the other block and the controls already in the regression; HAC(5) standard errors, bold marks $p < 0.05$, and n is the number of observations at the frequency named in the row. The controls differ from equation (2) in one respect in each gate. Gate 4 adds a one-day lag of the dependent variable to the market return and lagged log VIX, because daily gas returns are materially autocorrelated where defence returns are not; Gate 5 carries no market or volatility control, since neither is the relevant benchmark for a geopolitical-risk series, and conditions instead on six lags of its own outcome in levels and six in first differences. Gate 4 regresses daily Dutch TTF gas returns over windows (a) June 2021 to June 2022, (b) June 2022 to June 2023 and (c) June 2021 to December 2023. Gate 5 regresses the realised-conflict component of the Caldara and Iacoviello (2022) index on held-out days never used in sample, 637 at $h = 1$ and 629 at $h = 5$. All four Gate 4 cells estimated are reported.

that the apparatus finds effects where they exist.

Gate 4 also demonstrates how a significant result in this setting can dissolve under replication alone. Estimated exploratorily on an initial 81-day subsample of gas-crisis coverage, the build-up specification returned $p = 0.0028$ under the controls the confirmatory test would later use, and $p = 0.0020$ under an alternative control set — stable enough across specifications to appear robust. The pre-registered confirmatory estimate, on the same asset, specification, controls and calendar period, with nothing altered but the quantity of data, returned $p = 0.399$ on 222 continuous days. The result was not overturned by a better control or a stricter correction, but by fuller coverage of the period it had always concerned. This is the failure mode that pre-registration is designed to expose, and it is invisible to any robustness check conducted within the smaller sample.

Gate 5 removes prices from the test entirely and asks whether local perception anticipates the war's own escalation. If local media carry genuine information about the conflict, they should at minimum lead a measure of realised conflict, whether or not any asset prices it. On 637 held-out days never used in sample they do not: the local block returns $p = 0.21$ at one day and $p = 0.23$ at five, and remains insignificant when the outcome's own dynamics are added, as six lags in levels and six in first differences. This gate is weaker evidence than the other three, because its own Western control is significant at five days ($p = 0.033$) but not at one ($p = 0.171$); it is reported because a pre-registered test must be reported whatever it returns, and because a null on a non-market outcome closes off the possibility that the earlier nulls are artefacts of market microstructure.

4.3 Out-of-sample predictability

None of the preceding tests is a statement about forecastability, which requires separate evidence. A contemporaneous association may exist without being exploitable, and equally an absence of contemporaneous association does not establish that no predictive relationship exists at any horizon. Table 7 reports the out-of-sample results together with the power simulation that makes their null interpretable. The forecasting grid has five targets and ten predictors (attention share and tone for each of five ecosystems), giving 50 specifications. Every cell is nested and is arbitrated by the Clark–West test, for the reason given in Section 2.5.

Zero Clark–West rejections across 50 specifications is informative only to the extent the design could have rejected, and Panel B says how far that extends. On the three long-sample targets, an effect at the bottom of the 0.5–1.0% range Welch and Goyal (2008) and Campbell and Thompson (2008) treat as economically significant is detected in 81% of simulated paths, and one at the top in 98%. On the two Bloomberg targets the corresponding figures are 44% and 76%, so on those a meaningful effect would be missed more often than not and the null carries little weight. The best value actually observed, +0.10%, sits below even the 0.2% effect size that the design detects only about half the time — and its Clark–West p-value is 0.067, so not one of the fifty specifications reaches nominal significance, let alone survives correction. What can be claimed is therefore bounded in both directions: an effect large enough to matter economically would very probably have been seen, and an effect that escaped this design lies at or below the bottom of the range this literature treats as economically meaningful.

4.4 Discussion

The null is consistent across all four tests and both asset classes. Local perception adds nothing in coverage volume, in tone, in the anticipated/realised structure, in the asset with the most direct transmission channel, in a non-market outcome, or out of sample. The design is nonetheless demonstrably sensitive: the Western block is detected under multiple-testing correction in the anticipation test and at $p = 4 \times 10^{-5}$ in the gas test, and in the attention-and-tone test the local block itself is detected in seven of 31 cells before

Table 7: Out-of-sample return predictability and test power

Quantity		Value
Panel A: Clark–West results across the full grid		
Specifications (5 targets × 10 predictors)		50
Specifications with positive R^2_{OS}		3 of 50
Best R^2_{OS} observed	+0.0010	(+0.10%)
Smallest Clark–West p-value		0.067
Clark–West $p < 0.05$	**0**	(2.5 expected by chance)
Survive Benjamini–Hochberg (5% FDR)		**0**
Panel B: Simulated power, rejection rate at the 5% level		
True R^2_{OS} implanted	Long sample 1,855 OOS days *(3 targets)*	Bloomberg 686 OOS days *(2 targets)*
0.0% (nominal size)	0.02	0.02
0.2%	0.40	0.21
0.5%	**0.81**	**0.44**
1.0%	0.98	0.76
2.0%	1.00	0.97

Notes: Expanding-window, one-day-ahead forecasts with a minimum 250-day training window, re-estimated at each step. Each specification adds one perception predictor to a historical-mean benchmark, so the benchmark is nested and the comparison is arbitrated by the Clark–West test (Clark and West, 2007). R^2_{OS} is the Campbell–Thompson out-of-sample R^2 (Campbell and Thompson, 2008), the proportional reduction in mean squared forecast error relative to that benchmark. Out-of-sample length differs by target, and Panel B reports both: 1,855 days for the three free-data targets, and 686 for the two Bloomberg indices, which begin only in 2020. Panel B implants a predictor of known true R^2_{OS} into the corresponding return series, re-runs the identical expanding-window machinery over 1,000 paths per cell, and reports the share of paths in which Clark–West rejects. At 1,000 paths the standard error of each entry is at most 1.6 percentage points. Produced by `scripts/make_thesis_power.py`.
The two columns should not be read as one number. The power that justifies treating a null as informative is available on the long sample, where an effect of 0.5% is detected in 81% of paths. On the Bloomberg targets the same effect is detected in 44%, so for those two targets a true effect of that size would be missed more often than not, and the null there is weak evidence rather than a bound.

the tradeable lag is imposed. The informative feature of these results is their co-occurrence — a null on the local block in precisely the cells where an effect is detected on the Western block, or on the local block at the same-day alignment that a conservative reading cannot use. Media effects on prices are measurable in this design, and the local-language ones are not among those it measures.

One reading of the null is that Western investors respond to the signal that moved, and Section 3 identifies which signal that was. Section 3 showed that Russian state media's tone did not change when Russia invaded Ukraine, and that Ukrainian and Russian attention were already near saturation and had almost no room to rise. Western attention, by contrast, doubled in a day. If the marginal buyers and sellers of Western defence equities form their views from Western coverage, then they responded to the one series in the data that registered the event. That is an interpretation the results are consistent with rather than one they establish: the tests rule out an incremental local contribution, and do not identify what the Western contribution is. The information asymmetry between the two media environments is real and large; what does not follow is that the asymmetry advantages local media from the standpoint of a Western investor. Coverage already at saturation cannot rise further when an event occurs, and a series that does not move cannot explain one that does.

The result runs against the prior that motivated the design, which is stated here explicitly. Local media are closer to the events, cover them earlier and in far greater volume, and Bondarenko et al. (2024) had already established that local-language risk measures carry information English-language measures do not. The prior was therefore that local perception would contribute something, most plausibly through the anticipated/realised structure tested in Gate 3, and the pre-registrations were specified to detect such a contribution rather than to rule one out. Gates 4 and 5 were added because defence equities are a weak testbed for this question, the transmission from Russian-language reporting to a Western contractor's share price running entirely through Western investors. Neither the primary tests nor the two additional ones detect a contribution.

Reconciling that with the literature requires being precise about what Bondarenko et al. (2024) actually established. Their finding is that local-language geopolitical-risk shocks move the *Russian* economy while English-language shocks do not. The natural extrapolation — local sources are simply better measures of geopolitical risk — is the one this thesis rejects. What survives is narrower and more useful: the informative ecosystem is the one inhabited by the agents whose decisions are being measured. Russian firms and households read Russian newspapers, and Bondarenko et al. (2024) find that Russian coverage moves Russian output; the mirror proposition here — that Western fund managers read Western newspapers, and Western coverage is accordingly what these prices track — is one this evidence is consistent with rather than one it establishes, since the design identifies the local block's incremental contribution and not the Western block's own effect. Read that way the two results are not in tension at all, and neither is a claim about which journalists are better informed.

The results refine rather than contradict the defence-equity literature, and they sit more comfortably within it than a null might suggest. Covachev and Fazakas (2024) and Klomp (2025) document large abnormal returns on defence stocks at conflict onsets, and nothing reported here disputes that: the Western block is detected in the anticipation and gas tests, and Figure 1 shows the invasion repricing plainly. What this thesis adds is that the response is specific to the Western reporting layer rather than to conflict news as such — a distinction invisible to any design that aggregates coverage without regard to the publisher, as all of these studies do, and invisible for a practical reason, since separating publishers requires a translingual corpus available only for part of the period they cover.

The finding that continuous news measures do not predict returns is, if anything, the established position rather than a departure from it. Apergis et al. (2018) reach the same conclusion for returns on 24 defence firms using the published geopolitical-risk index and a nonparametric design far removed from the one used here. Where their result and this one appear to diverge is on volatility: they find geopolitical risk predicting the volatility of about half their firms, whereas Section 4.4 reports that war variables fail to improve on HAR and GARCH benchmarks. The two are compatible, because the questions differ. Theirs is whether geopolitical risk carries in-sample information about volatility; the one asked here is whether war-specific news adds *out-of-sample* forecasting accuracy to volatility models that already exploit the persistence of realised variance. A variable can satisfy the first and fail the second, and conflict news evidently does.

The physical record of the war does not forecast defence-equity returns either, which closes off the reading that the news measures are simply a poor proxy for the war itself. The design is an expanding-window forecasting race on the September 2022 – June 2026 subsample where the Ukrainian Air Force attack data exist. Five information sets are compared: financial controls alone (F), those plus physical attack variables (P), plus news variables (N), both (PN), and both plus the pairwise tone gaps between ecosystems (PNG). Each is fitted by ridge regression and by gradient boosting, and each is scored against a historical-mean and an AR(1) benchmark, over three targets, two horizons and two samples — twelve cells carrying twelve fitted models each, or 144 specifications. Sixteen achieve a positive out-of-sample R^2 and none survives Benjamini–Hochberg correction. Table 9 in the appendix reports the counts by information set: adding physical attack data to the financial controls raises the number of cells with positive R^2_{OS} from one to four, and none of those four is significant. Realised physical intensity is therefore no more informative about subsequent defence-equity returns than perceived intensity is.

The volatility exercise points the same way but is the weakest evidence in this thesis, and the reason is the volatility measure. Returns are close to unforecastable in efficient markets whereas volatility is persistent, so conflict information carrying no signal for the conditional mean might still carry one for the conditional variance, which makes this the specification where a positive result was most plausible. **Realised variance here is the sum of squared daily returns over the forecast horizon, dated at the start of that window, because intraday prices were not available for these indices.**

That is a far noisier proxy than the intraday realised variance for which the HAR model was designed (Corsi, 2009), and the cost is visible: the plain HAR-RV benchmark attains a mean QLIKE of 1.93 against 0.95 for GARCH, 0.96 for GJR-GARCH and 0.98 for EGARCH, so on this proxy the HAR family is roughly twice as inaccurate as the GARCH family it is normally competitive with.

Within that limitation the result is clean, and it is reported as a limited one. Forty-eight augmented HAR-RV-X specifications — log realised variance on its own one-, five- and twenty-two-day lags, plus one of the four war blocks, estimated by ridge on an expanding window — are compared against the HAR-RV benchmark they nest, under QLIKE. None improves significantly on it, eight are nominally significantly worse, and the six that survive multiple-testing correction are all deteriorations; Table 10 in the appendix gives the breakdown. Augmenting a volatility model with war variables degrades its accuracy on this measure rather than improving it. What cannot be concluded is that conflict news carries no information about defence-equity volatility, because a test built on squared daily returns has limited power to detect it. Both exercises in this subsection are exploratory rather than pre-registered.

Six specifications in this setting yield significance under defensible initial choices and lose it under stricter ones, and the six mechanisms are distinct. Two fail on an omitted variable: a defence-volatility result and a build-up threat result, each estimated with the S&P 500 as the market control for European defence equities, and each disappearing when the STOXX 600 replaces it, the second moving from $p = 0.0001$ to $p = 0.84$. One fails on sample truncation: the Gate 3 grid clears its pre-registered rule on a partial sample and fails once the held-out window is included, seven surviving cells becoming two. One fails on replication with more data: the gas specification, $p = 0.0028$ on 81 days and $p = 0.399$ on 222 days of the same calendar period. One fails out of sample: an escalation result significant in both halves of the in-sample period and $p = 0.21$ on held-out data. One fails on corpus composition: the state-versus-independent tone difference, which depends on outlets that the ownership rule of Section 2, correctly applied, places outside the Russian independent block; each such reassignment moves the difference *further* from significance, and on the register as specified it stands at $p = 0.561$.

Taken together these six bound how much confidence a single significant specification warrants in this literature. Each is significant at conventional levels, each admits a plausible economic mechanism, and each survives at least one robustness check before failing another. Two are eliminated by a better control, one by additional data, one by a rule fixed before the data existed, one by holding data back, and one by correcting a classification error — and in every case the eliminating test is one that a study reporting the positive would have had no particular reason to run. They are reported here because in a literature whose regressors are constructed from text, the set of choices under which a result survives is part of the result, and a null established against this background is considerably more informative than one obtained without it.

5 Conclusion

Local coverage provides no robust incremental information about defence-equity returns beyond what Western coverage already carries. That is the answer this thesis can defend, and it is a null rather than a reversal: perception from Ukrainian outlets, Russian state media and Russian independent outlets adds no explanatory power conditional on Western coverage, in any of four tests, across two asset classes, and against one non-market outcome. The null is not attributable to a design unable to detect media effects: the Western block is detected under correction in the anticipation and gas tests, and in the attention-and-tone test the local block itself is detected at the same-day alignment and lost at the conservative lagged one. A power simulation establishes that on the three long-sample targets, with 1,855 out-of-sample days each, effects in the range the predictability literature treats as economically meaningful would have been detected in 81% to 98% of simulated paths. None is observed. The claim is that such an effect is unlikely to have been missed on those targets, not that it has been excluded, and it does not extend to the two Bloomberg indices, where the same simulation returns 44% and 76%.

This inverts the asymmetry Bondarenko et al. (2024) document on the other side of the conflict. For the Russian domestic economy local coverage carries information that English-language coverage does not; for Western defence equities local coverage carries nothing beyond it. The evidence here is more consistent with Western than with local media being the relevant information set for these assets, but it does not establish that Western coverage is priced, and the stronger claim should not be read into the weaker one. Neither study supports a general statement about whether local media matter. Together they suggest that the question is empirical for each market, and that the answer tracks whose readers hold the asset.

The practical implication extends beyond this conflict. Multilingual geopolitical-risk indicators are constructed on the premise that sources closer to an event carry information Anglophone coverage omits. Tested in close to the most favourable setting available, that premise does not hold: a Ukrainian- or Russian-language risk index constructed for the purpose of trading Western defence equities measures a signal those prices do not respond to. The premise is not refuted in general, since Bondarenko et al. (2024) establish it for the Russian domestic economy. What follows is that it cannot be assumed. Whether local-language sources add information is an empirical question about the distance between an information ecosystem and a market's marginal investor, and it must be settled market by market rather than in general.

One limitation is specific to how coverage is measured, and it cuts in the direction of the result. The coverage-share variable counts articles by whether they name a Ukrainian or Russian place, which for a Western outlet is close to a conflict filter but for a Ukrainian or Russian one is close to a domestic-news filter. The local series therefore sit at 70–86% of output and have little room to rise when the conflict escalates, so part of what the tests read as an absent local signal may be a ceiling in the measure rather than an absence in the world. Two things bound how much of the result this can explain. The anticipated/realised variables tested in Gate 3 are built from theme codes rather than locations, are conflict-specific by construction, and return the same null. And the local series are not in fact static: Panel D of Table 2 shows their daily changes are two to four times as variable as the Western ones, so there is movement for a test to find. Replacing the location proxy with a theme-based filter throughout would settle it, at a query cost roughly eight times higher.

Three further limitations bound what this thesis can claim. The first is that the analysis is associational rather than causal: coverage and prices respond to the same underlying events, and the conditional design separates ecosystems from one another but does not identify a causal effect of coverage on prices. The second is measurement: GDELT tone is a dictionary-based score, and Loughran and McDonald (2011) give good reason to treat such scores as indicators of coverage negativity rather than of investor belief; multilingual transformer models applied to article text would sharpen the signal, though they would not change the classification question this thesis is about. The third is frequency, and it is the one that most limits what the Gate 2 timing contrast can be made to mean. A GDELT day is a full UTC day, so a same-day aggregate mixes coverage published before the European close with coverage published after it, and lagging that aggregate shifts the whole bundle rather than separating the two. The lagged specification is therefore the conservative one, in that everything in it precedes the market open, but it does not identify when within the day information arrived or was impounded. The fall from seven surviving cells to one is consistent with same-day news being priced within the trading day, with a mechanical artefact of matching a return to coverage partly published after it, and with information that decays inside a day; daily aggregates cannot distinguish them. Only timestamped intraday data could.

One apparent limitation can be discharged directly. An index-level null may in principle mask cross-sectional variation, with heavily war-exposed firms responding where a diversified index does not. A firm-level panel of 31 defence names and 85,065 firm-days, with exposure measured from published SIPRI arms-revenue data, produces no exposure gradient in any war window: nominal significance appears only in the years before the full-scale invasion, and in a full sample of which 59% falls in those years, and none of the ten cells survives correction. The index-level result is therefore not an artefact of aggregation.

The extension of greatest value would establish whether this result generalises. Applying the same ecosystem comparison to other conflicts, such as the Hamas–Israel war or tensions in the Taiwan Strait, would distinguish a general property of defence-equity pricing from a feature specific to this war, in which the local-language ecosystems were already at coverage saturation before its defining event. A second extension would move to intraday prices, at which the anticipated-versus-realised distinction that Gate 3 cannot resolve at daily frequency becomes observable. A third would apply the design to assets whose marginal investors are not Western — rouble-denominated instruments, or the sovereign debt of states bordering the conflict — which constitutes the sharpest available test of the proposition that each market prices the information ecosystem its own participants inhabit.

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A Appendix

The tables below support the descriptive section and the two exploratory forecasting exercises of Section 4.4, and are placed here rather than in the body to keep the number of exhibits in the paper itself to ten.

Table 8: Correlations between media ecosystems and defence-equity returns (daily first differences)

	Between media ecosystems					With defence returns	
	UA	RU_ST	RU_IN	WEST	EN_GL	WAERLST	BSHIELDT
Panel A: change in coverage share							
Ukrainian (UA)	1.00					−0.01	0.01
Russian state (RU_ST)	0.47	1.00				0.02	0.02
Russian independent (RU_IN)	0.05	0.13	1.00			−0.03	−0.02
Western (WEST)	0.13	0.16	0.15	1.00		−0.03	−0.03
Native-English (EN_GL)	0.02	0.06	0.09	**0.60**	1.00	−0.00	−0.04
Panel B: change in coverage tone							
Ukrainian (UA)	1.00					−0.00	−0.02
Russian state (RU_ST)	0.39	1.00				0.03	0.05
Russian independent (RU_IN)	0.35	**0.57**	1.00			0.03	0.04
Western (WEST)	0.36	0.35	0.33	1.00		0.02	−0.01
Native-English (EN_GL)	0.26	0.28	0.28	**0.58**	1.00	0.02	0.01

Notes: Pearson correlations of daily first differences, the transform used throughout. Correlations are pairwise: the media block uses the 3,072 estimation-sample days with a lagged value available, the two right-hand columns the days on which the relevant Bloomberg index also trades (936 for WAERLST, 921 for BSHIELDT). Only the lower triangle is shown; the matrix is symmetric.

Table 9: Return forecasting race by information set

Set	Contents	Specs	$R^2_{\mathrm{OS}} > 0$	Best R^2_{OS}	BH surv.
F	Financial controls only	24	1	0.019	0
P	F + physical attack variables	24	4	0.073	0
N	F + news variables	24	1	0.033	0
PN	F + physical + news	24	3	0.060	0
PNG	PN + pairwise ecosystem tone gaps	24	4	0.059	0
Benchmarks					
	Historical mean	12	0	0.000	—
	AR(1)	12	1	0.009	—

Notes: Expanding-window one-step-ahead forecasts of daily returns on WAERLST, BSHIELDT and ITA, at horizons of one and five days, on the full sample and on the September 2022 – June 2026 attack subsample. Each information set is fitted by ridge regression and by gradient boosting, giving 24 specifications per set across the twelve target–horizon–sample cells and 144 rows in total with the benchmarks. R^2_{OS} is measured against the historical mean, and Benjamini–Hochberg correction is applied to the Clark–West p-values across the grid. Exploratory, not pre-registered.

Table 10: Volatility forecasting race, augmented HAR-RV-X against HAR-RV

Model	Specs	Mean QLIKE gain (%)	Sig. better	Sig. worse	BH surv.
HAR-RV-X (N)	12	−4.4	0	1	0
HAR-RV-X (P)	12	−65.0	0	2	2
HAR-RV-X (PN)	12	−62.6	0	2	2
HAR-RV-X (PNG)	12	−64.8	0	3	2
Mean QLIKE by model class, lower is better					
GARCH 0.95	GJR-GARCH 0.96	EGARCH 0.98	**HAR-RV 1.93**		

Notes: Realised variance is the sum of squared daily returns over the forecast horizon, dated at the start of that window; intraday prices were not available for these indices. HAR-RV regresses log realised variance on its own one-, five- and twenty-two-day lags, estimated by ridge on an expanding window and exponentiated back; HAR-RV-X adds one war block (P, N, PN or PNG). Models are evaluated over the same twelve cells as Table 9. Gains are relative to HAR-RV, which the augmented models nest; significance is at 5% before correction, and "BH surv." counts cells surviving Benjamini–Hochberg.
The final row is the limitation, not an aside. On a squared-daily-return proxy the HAR family is roughly twice as inaccurate as the GARCH family it is normally competitive with, so this exercise has limited power and its null bounds only what such a proxy can detect.